

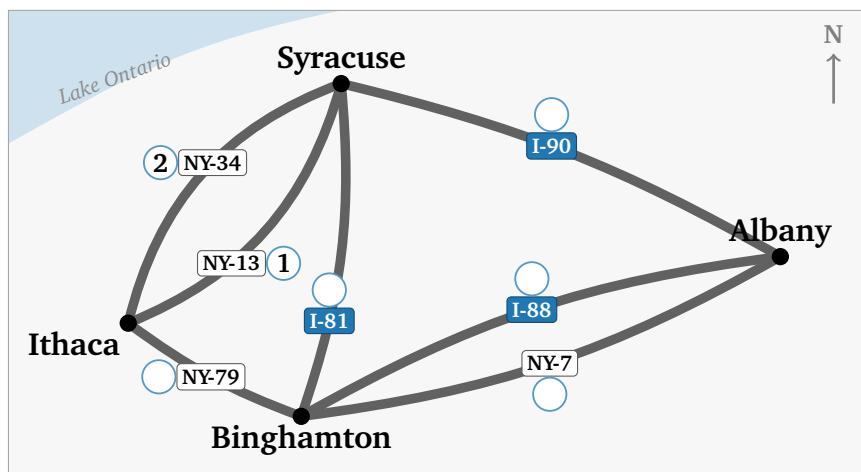
# Drive Every Highway Once, Then Teach a Computer to Do It

Do this before the lecture, in pencil. Nothing here needs a word you have not been given on the sheet.

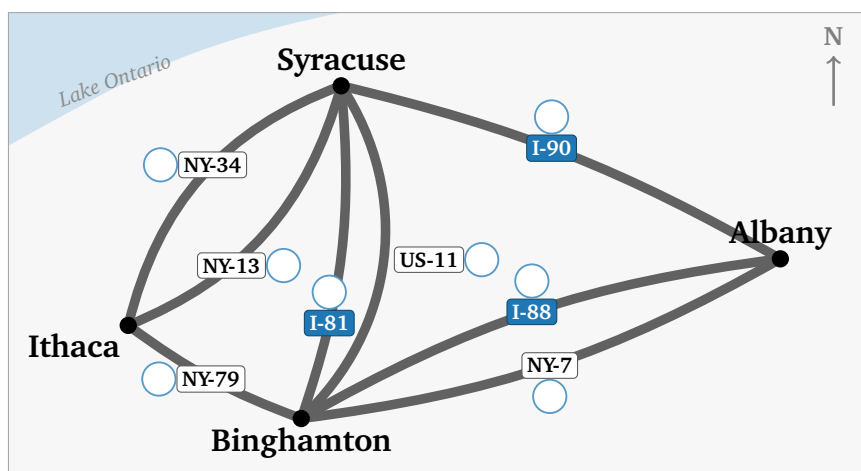
## Part 1 — Seven highways

Seven highways join four Upstate New York cities: Ithaca, Syracuse, Binghamton, Albany.

**Question 1(a):** Drive every highway exactly once, without lifting your pencil. Start and finish anywhere; cities may repeat, highways may not. Write in each circle when you drive that road — 1 first, then 2, on to 7. Two are filled in: you start at Ithaca, take NY-13, then NY-34 straight back.



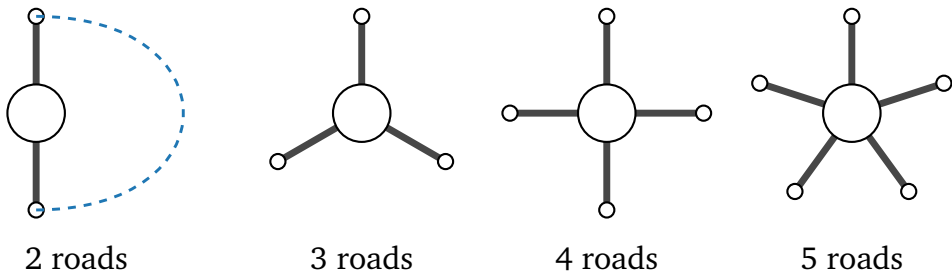
**Question 1(b):** The same four cities, and an eighth highway: US-11, which really does run alongside I-81. Number your route again.



Can you drive every highway without lifting your pencil? \_\_\_\_\_

**Question 2:** One city, on its own. Arriving uses up one highway, leaving uses up another — so highways get used in **pairs**.

Join two road-ends with a curved line to mean “in here, out there”. Pair up as far as you can. The first is done.



|                 | 2 roads | 3 roads | 4 roads | 5 roads |
|-----------------|---------|---------|---------|---------|
| pairs you drew  | 1       |         |         |         |
| roads left over | 0       |         |         |         |

- (a) Which ones can you drive straight *through*? \_\_\_\_\_
- (b) What do those numbers share? \_\_\_\_\_
- (c) A left-over road has no partner, so this city must be the tour’s \_\_\_\_\_

**Question 3:** Count the highways at each city. Ithaca is done.

|                | I | S | B | A | how many have a left-over? |
|----------------|---|---|---|---|----------------------------|
| seven roads    | 3 |   |   |   |                            |
| with US-11 too |   |   |   |   |                            |

One start, one finish. So at most \_\_\_\_\_ cities may have a left-over road.

Seven roads: \_\_\_\_\_ Eight roads: \_\_\_\_\_

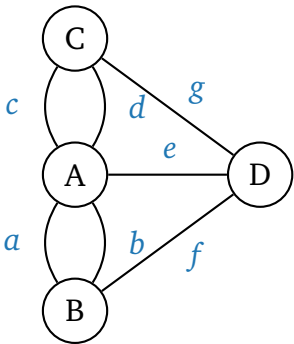
No route with eight because \_\_\_\_\_

**Question 4:** The same puzzle, 1736. Königs-berg: four pieces of land, seven bridges. Two pairs join the same land; both of a pair must be crossed.

Bridges: A \_\_\_\_\_ B \_\_\_\_\_ C \_\_\_\_\_ D \_\_\_\_\_

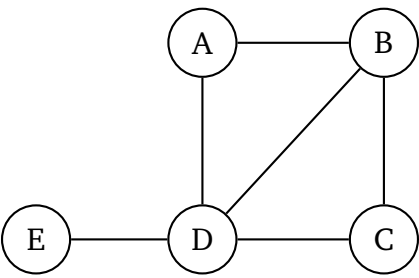
Is the walk possible? \_\_\_\_\_

Destroying one bridge is enough. Which ones work? \_\_\_\_\_



Part 2 — Three kinds of route

A different network: five bus stops.



Question 5(a): Fill in the table. The last column takes one of these three names. First row done.

Repeats nothing: **path**.    Never repeats a line: **trail**.    Repeats both: **walk**.

| route     | repeats a line? | repeats a stop? | name |
|-----------|-----------------|-----------------|------|
| A B D B C | yes (B–D twice) | yes (B)         |      |
| A B D C B |                 |                 |      |
| A B C D E |                 |                 |      |

Question 5(b): A route from A to E that repeats a stop but no line:

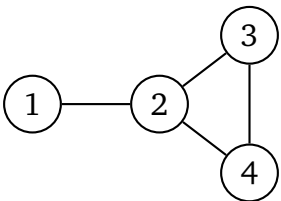
A \_\_\_\_\_ E

Question 5(c): Back to the highway map. Your drive in 1(a) used no road twice, but it did come back to a city it had already visited. It is a \_\_\_\_\_.

Question 5(d): It could not have been a path. A path of 7 roads would have to touch \_\_\_\_\_ different cities, and the highway map has only \_\_\_\_\_.

Part 3 — Predictions for the notebook

Part 4 sends you to a notebook that will run this network for you. Work the answers out on paper first; the notebook will check them.



**Question 6(a):** Write **1** in row  $i$ , column  $j$  if a line joins  $i$  and  $j$ , else **0**. Row 1 done.

$$A = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 0 & 0 \\ 2 & & & & \\ 3 & & & & \\ 4 & & & & \end{array}$$

**Question 6(b):** Add up each row, then compare with the picture.

row totals: \_\_\_\_\_

Each row total is the \_\_\_\_\_

of that node.

**Question 6(c):** A **2-step route** goes along one line, then along one more. From node 1 to node 3,  $1 \rightarrow 2 \rightarrow 3$  is one such route. Using the picture, count the 2-step routes:

from 1 to 3: how many? \_\_\_\_\_ write one out:  $1 \rightarrow \_\_\_\_\_ \rightarrow 3$

from 1 to 2: how many? \_\_\_\_\_ why so few? \_\_\_\_\_

from node 2 back to node 2: how many? \_\_\_\_\_

**Question 6(d):** The notebook in Part 4 will square your grid for you. Before it does, say what you expect to find in  $A^2$ .

row 1, column 3: \_\_\_\_\_ row 1, column 2: \_\_\_\_\_

the diagonal: \_\_\_\_\_

## Part 4 — The lab, on your own



**Do this one by yourself**, with this sheet beside you. The notebook takes the same four cities and seven highways you have been drawing on, and asks you to write them down in a form a computer can read.

**Point a camera at the square.** If that fails, type [go.skojaku.com/m01lab](http://go.skojaku.com/m01lab)